Computer Engineering Department

National Institute of Technology, Kurukshetra-136119

Mid Term 2 , Even Semester 2024-25

Paper: Image Processing (CSPE 306)

Max Marks:15 **Total time: 50 m**

Q1. An imaging system captures the following intensity values for a grayscale object and its background: Average intensity of the object f-object=180, Average intensity of the background f-background=120, Maximum image intensity f-max=200, Minimum image intensity f-min=80

(a) Calculate the Weber contrast using the given values.

(b) Calculate the Michelson contrast using the maximum and minimum intensities. **[3 marks]**

Q2. What would be the impact of removing the LSB (Least Significant Bit) of the pixel values in the histogram of the following image?

$$\begin{bmatrix} 6 & 7 & 1 \\ 5 & 1 & 4 \\ 1 & 2 & 3 \end{bmatrix}$$

[2 marks]

Q3. An 8-bit grayscale image has intensity values ranging from 0 to 255. You are asked to apply a piecewise linear transformation defined as follows: **[4 marks]**

$$T(r) = \begin{cases} 0.5r & \text{for } 0 \leq r < 100 \\ r & \text{for } 100 \leq r < 200 \\ 0.2r + 100 & \text{for } 200 \leq r \leq 255 \end{cases}$$

(a) Compute the output intensity values for the following input pixels: r=50,120,220.

(b) Explain how this transformation affects the contrast in dark, mid-tone, and bright regions of the image.

Q4.

(a) Consider a 3×3 high-pass filter mask defined as:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

1. What type of spatial features does this filter enhance?
2. Derive the equivalent filter H(u,v) in the frequency domain. **[4 marks]**

(b) What would happen if low or high pass filters are repeatedly applied to an image? Is this repeated operation helpful in any way? **[2 marks]**